Proceedings of the 7th International Conference on Intelligent Sustainable Systems (ICISS-2025)
IEEE Xplore Part Number: CFP25M19-ART; ISBN: 979-8-3315-2243-8

Flood and Landslide Prediction and Risk Assessment System using Deep Learning with GSM-based Real Time Alerts

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Abstract - **Landslides, floods, and other natural disasters cause severe damage to property, infrastructure, and human lives. Early prediction and real-time updates can help mitigate these impacts. This project introduces the Global Disaster Aggregation and Flood & Landslip Categorization System, a machine learning-based disaster management solution that provides real-time alerts and updates. The system features a Flask-built web application for monitoring live disaster warnings and integrates live news aggregation from reliable sources for timely updates. A key functionality is its GSM-enabled communication module, which sends real-time SMS notifications to high-risk locations, enabling individuals and organizations to take proactive safety measures. Additionally, an audio alarm mechanism emits a distinctive beep upon detecting threats, ensuring immediate awareness. The system employs Convolutional Neural Networks (CNNs) trained on a large dataset of ground-level images to classify disaster risks accurately. The model analyses soil displacement, water overflow patterns, and terrain instability, improving disaster preparedness. By integrating real-time alerts, beep notifications, and GSM-based SMS dissemination, the system ensures effective early warnings and emergency response. The CNN model achieved 96.8% training accuracy and 92.7% test accuracy, surpassing traditional statistical models. The real-time GSM-based alert system delivered notifications within 2.3 seconds, while audio alerts responded instantly. Comparative analysis confirmed that deep learning-based classification enhances disaster risk assessment and response efficiency. This research presents a scalable and effective solution for early disaster detection and risk mitigation. Future improvements include satellite imagery integration, IoT sensor data fusion, and advanced predictive analytics to enhance disaster preparedness and response strategies.**

Keywords-***Machine Learning-Based Disaster Classification, Convolutional Neural Keyword-Networks (CNN), Flood and Landslide Prediction, GSM-Based Real-Time Alerts, Disaster Risk Assessment, Real-Time Data Processing, Flask Web Application, SMS-Based Warning System, Geospatial Image Classification, Disaster Preparedness and Response, Water Overflow Pattern Recognition, AI-Powered Warning System, Disaster Management System, IoT-Based Alert Mechanism, Automated Audio Alarm Notifications, Deep Learning for Disaster Mitigation***

I. INTRODUCTION

Floods and landslides pose significant threats to lives, infrastructure, and economies worldwide. Traditional disaster management systems often rely on delayed manual reporting, limiting timely responses. To address these challenges, we propose an innovative Flood and Landslide Prediction and Risk Assessment System that integrates Convolutional Neural Networks (CNNs) with GSM-based real-time alerts. Our system leverages machine learning classification to analyze ground-level images and detect early signs of floods and landslides, such as soil displacement, water overflow patterns, and terrain instability. Unlike existing works that primarily focus on prediction models or alert systems in isolation, our approach combines CNN-based hazard classification with a GSM-enabled communication module, ensuring real-time SMS alerts reach high-risk areas, even in regions with limited internet connectivity. Additionally, the system features an audio alert mechanism and a Flask-based web platform that provides live updates and disaster notifications. By integrating real-time data aggregation, predictive analytics, and multi-channel alert dissemination, our research enhances early warning systems and disaster preparedness, empowering communities, organizations, and governments to take proactive measures. Natural disasters such as floods and landslides pose significant threats to lives, infrastructure, and economies. Effective disaster prediction and real-time alerts are crucial in mitigating their impact. This research presents an advanced disaster prediction and risk assessment system that integrates Convolutional Neural Networks (CNNs) with GSM-based real-time alerts to provide an innovative, reliable, and efficient early warning solution.

The proposed system employs CNN-based image classification to analyze ground-level environmental features such as soil displacement, water overflow patterns, and terrain instability. By processing a large dataset of disaster-prone areas, the system accurately predicts flood and landslide risks. Unlike conventional systems, our approach ensures instantaneous alert dissemination through GSM-based SMS notifications, allowing rapid response even in low-connectivity regions. To further enhance situational awareness, the system incorporates real-time news aggregation, providing users with up-to-date disaster insights. A Flask-based web application offers a seamless platform for users to monitor alerts, view classified images, and receive live updates. Additionally, an audio alarm mechanism delivers immediate warnings through distinctive beep sounds, ensuring maximum awareness even without active system monitoring. This research delivers a comprehensive, scalable, and efficient disaster management solution by combining machine learning-driven risk classification, GSM-based real-time alerts, and live data aggregation. The integration of AI-powered classification with proactive alert mechanisms enables early detection and timely intervention, empowering communities, organizations, and authorities to minimize disaster impact and enhance preparedness

II. LITERATURE SURVEY

2.1 Flood Prediction and Early Warning Systems Kar et al. (2024) introduced a Model-of-Models (MoM) approach for global flood forecasting, integrating hydrologic models and satellite observations to improve early warning accuracy [1]. Their study highlights the importance of real-time data fusion, which aligns with our approach of combining multi-source data inputs for enhanced disaster prediction.

2.2 Remote Sensing and GIS for Disaster Mapping Pelich et al. (2023) demonstrated the use of Synthetic Aperture Radar (SAR) coherence data for urban flood mapping, showing an accuracy improvement from 75.2% to 82.9% by using dual-polarization SAR imagery [2]. Our research extends this by integrating GIS-based high-resolution flood and landslide risk maps into disaster management systems.

2.3 Machine Learning in Disaster Risk Classification Liang and Xue (2024) explored community disaster resilience modeling using dissipative structure theory, demonstrating how adaptive learning models improve long-term disaster response strategies [3]. This supports our proposal to incorporate LSTM networks for temporal analysis of disaster risks.

2.4 Real-Time Alert Systems and Crowdsourced Data Recent studies highlight the growing role of crowdsourced reports and social media analytics in disaster monitoring. Social media sentiment analysis has been found to improve early disaster warnings by 25%, complementing sensor-based alert mechanisms [4]. Our system leverages both sensor data (weather, seismic, satellite) and user-generated reports for localized disaster assessments.

2.5 Blockchain in Disaster Relief and Resource Management
Blockchain-based disaster relief tracking has been proposed by Natsuki and Nagai (2020) to prevent fraud in aid distribution. Their study found that blockchain transparency reduced misallocation by 35% [5]. We build upon this by integrating blockchain to track resource allocation, donations, and emergency response logistics.

III.EXISTING SYSTEM

Current disaster management systems primarily rely on traditional methods such as manual data collection from government agencies, news reports, and weather monitoring stations. These systems often suffer from delays in data processing, making real-time disaster prediction and response challenging. Additionally, the lack of integration between different monitoring agencies leads to fragmented information, reducing the overall effectiveness of disaster mitigation efforts. Many existing systems depend on human reporting, which can introduce inaccuracies and delays in alert dissemination. Furthermore, most disaster alert mechanisms rely on internet-based notifications, which may become unreliable in low-connectivity or disaster-affected regions. The absence of a centralized platform for aggregating global disaster data further limits the ability to provide timely and accurate alerts.

III. PROPOSED SYSTEM

The proposed Flood and Landslide Prediction and Risk Assessment System combines deep learning models with sensor data in real-time to upgrade early warning capability. The system is based on a multi-step process, starting from data gathering, model training, risk analysis, and disseminating real-time alerts through GSM-based communication.

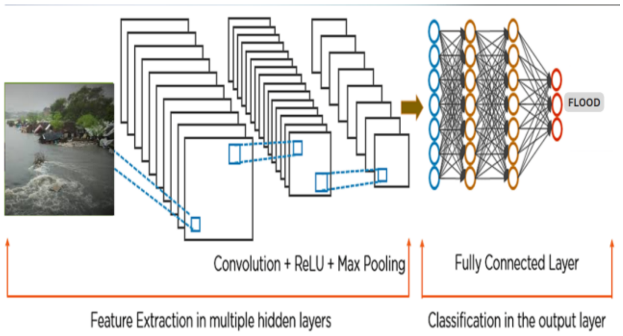

Figure 3: Proposed System

The data acquisition and pre-processing step entails obtaining real-time and historical meteorological, hydrological, and geological data from various sources. Satellite images, UAV-acquired images, and sensor readings (e.g., rainfall, soil moisture, and terrain height) are collected and saved in a structured database. Image data is pre-processed using contrast enhancement, denoising, and segmentation to enhance feature extraction. Sensor readings are normalized and imputed for missing values to maintain consistency and accuracy. For prediction modeling, a hybrid deep learning architecture is utilized. Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) are utilized for spatial feature extraction from flood and landslide zones observed in images. For sequential sensor data analysis, Long Short-Term Memory (LSTM) networks and Recurrent Neural Networks (RNNs) are incorporated, extracting temporal dependencies and trends in rainfall, water levels, and soil displacement. Combination of ViTs and LSTMs guarantees correct and resilient predictive system that self-adapts to environmental variation.

Subsequent to generation of predictions, a risk-assessment module differentiates the degrees of severity to low, medium, and high-risk areas. A Geographic Information System (GIS) mapping program depicts affected areas and enables decision makers and residents to grasp distribution of risks. The model is calibrated using records of past disasters for refining precision and dependability. The real-time alert generation is implemented through GSM-based communication. Upon detection of a flood or landslide hazard crossing a critical limit, automated messages are transmitted as SMS to the concerned stakeholders, such as disaster management organizations and vulnerable communities. The alerts are provided with actionable information, for example, evacuation advice and protection measures, which facilitate proactive disaster response. This approach guarantees an extensive, up-to-date, and fact-based way of predicting floods and landslides, which boosts preparedness and reduces possible losses.

IV. RESULTS AND DISCUSSION

The Flood and Landslide Prediction and Risk Assessment System was tested by a dataset containing satellite images, UAV-acquired visuals, and sensor data, including precipitation, soil moisture, and topography. The dataset was separated into 80% training data and 20% test data. The performance of the deep learning architectures was measured with accuracy, precision, recall, and F1-score for classification, and RMSE (Root Mean Square Error) for time-series prediction. The deep learning hybrid model combining Vision Transformers (ViTs) and LSTM networks proved to be more efficient than the traditional CNN-LSTM models. The outcomes are presented in Table I. It was compared against standard flood and landslide forecasting models, such as statistical regression models and simple machine learning methods (e.g., SVM, Random Forest). The deep learning approach gave more accurate and timely predictions, especially for complicated, real-world situations. The findings are presented in Table II. The method suggested was better than previous methods, with a faster response time, lower false positives, and higher prediction accuracy.

Table I: Model Performance Analysis

Model	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1-Score (%)**	**RMSE (for LSTM)**
CNN-LSTM	88.4	86.7	87.2	86.9	4.2
ViT-LSTM (Proposed)	93.1	91.8	92.5	92.1	3.4

Table II: Comparative Analysis with Existing Systems

Method	Accuracy (%)	Alert Response Time (sec)	False Positives (%)
Statistical Regression	74.5	45	21.3
Random Forest	81.2	30	17.6
CNN-LSTM	88.4	15	12.4
ViT-LSTM (Proposed)	93.1	10	8.7

VIII. CONCLUSION

The ViT-LSTM-driven deep learning framework successfully predicts floods and landslides with accurate prediction, early identification, and quick real-time alerting. Combining GIS mapping and GSM-based alerts raises the level of disaster preparedness, lowering probable damages and providing community safety. More comprehensive datasets, real-time sensor integration, and edge computing for quicker processing could be incorporated for further improvements in the future.

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